these arguments there was a series of paradoxes that became celebrated as ‘Zeno’s paradoxes’; they seek to show how all appearance is illusory, arguing that the commonplace notion of motion is absurd.[12]

The most famous of Zeno’s paradoxes is presented in the form of a brief fable: the tortoise challenges Achilles to a race, starting out with a ten-metre advantage. Will Achilles manage to catch up with the tortoise? Zeno argues that rigorous logic dictates that he will never be able to do so. Before catching up, in effect, Achilles needs to cover the ten metres, and in order to do this he will take a certain amount of time. During this time, the tortoise will have advanced a few centimetres. To cover these centimetres, Achilles will have to take a little more time but, meanwhile, the tortoise will have advanced further, and so on, ad infinitum. Achilles therefore requires *an infinite number of such times* to reach the tortoise, and an *infinite number of times*, argues Zeno, is *an infinite amount of time*. Consequently, according to strict logic, Achilles will take an infinite time to reach the tortoise; or rather, we will never see him do so. Since, however, we do see the swift Achilles reaching and overtaking as many tortoises as he likes, it follows that what we see is irrational, and therefore illusory.

Let’s be honest: this is hardly convincing. Where does the error lie? One possible answer is that Zeno is wrong because it is not true that by accumulating an infinite number of things one ends up with an infinite thing. Think of taking a piece of string, cutting it in half, and then again in half, and again, ad infinitum. At the end, you will obtain an infinite number of small pieces of string; the sum of these, however, will be finite, because they can only add up to the length of the original piece of string. Hence, an infinite number of strings can make a finite string; an infinite number of increasingly short times may make a *finite* time, and the hero, even if he will have to cover an infinite number of distances, ever smaller, will take a *finite* time to do so, and will end up catching the tortoise.

It seems that the paradox is resolved. The solution, that is, is in the idea of the continuum – arbitrarily small times may exist, an infinite number of which make up a finite time. Aristotle is the first to intuit this possibility, which was subsequently developed by ancient and modern mathematics.[fn1]

But is this really the correct solution in the *real* world? Do arbitrarily short strings really exist? Can we really cut a piece of string an *arbitrary* number of times? Do infinitely small amounts of time exist? This is precisely the problem that quantum gravity will have to face.

According to tradition, Zeno had met Leucippus and had become his teacher. Leucippus was therefore familiar with Zeno’s riddles. But he had devised a *different* way of resolving them. Maybe, Leucippus suggests, nothing arbitrarily small exists: there is a lower limit to divisibility.